



ReliSource ML Engineer Interview – Process & ML Core Theory



ReliSource Interview Process (What to Expect)

Based on Glassdoor reports and community data from May 2024:

Step-by-Step Process

Stage	Format	Focus
Stage 1	Written/Online Exam (1–2 hrs)	ML fundamentals, problem solving, Python/SQL
Stage 2	Kaggle-style ML Competition (on-site)	Build & optimize a model on a given dataset
Stage 3	Technical Interview Round 1	ML concepts, algorithms, your projects
Stage 4	Final Discussion with Team Lead	Advanced ML, system design, communication

 **Key Insight from 2024 Glassdoor report:** ReliSource hosted an **on-site Kaggle competition** as Stage 1 for ML roles. Top performers from that competition moved to interview rounds. The interviews focused on: **overfitting, underfitting, data imbalance, bias-variance tradeoff, k-fold cross-validation**, and **fundamental ML problem-solving skills**.



Section 1: Core Machine Learning Q&A

Q1. What is the bias-variance tradeoff?

Answer:

- **Bias** = error due to overly simplistic assumptions. High bias → underfitting (model misses patterns).
- **Variance** = error due to sensitivity to small fluctuations in training data. High variance → overfitting (model memorizes noise).

The tradeoff: reducing bias often increases variance and vice versa. The goal is to find the sweet spot that minimizes **total error**:

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$$

Example: A linear model on non-linear data has high bias. A deep decision tree without pruning has high variance.

Q2. What is overfitting and how do you prevent it?

Answer:

Overfitting occurs when a model performs well on training data but poorly on unseen data — it has "memorized" the training set instead of learning the underlying pattern.

Prevention techniques:

- **Regularization** (L1/Lasso, L2/Ridge) — penalizes large weights
 - **Dropout** — randomly disables neurons during training (DL)
 - **Cross-validation** — use k-fold CV to estimate generalization
 - **Early stopping** — stop training when validation loss increases
 - **More data** — more training examples reduce overfitting
 - **Simpler model** — reduce model complexity
 - **Data augmentation** — artificially increase dataset diversity
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Q3. What is underfitting and how do you fix it?

Answer:

Underfitting occurs when a model is too simple to capture the pattern in data. High training error and high validation error.

Fix:

- Use a more complex model
 - Add more relevant features (feature engineering)
 - Reduce regularization strength
 - Train for more epochs (DL)
 - Remove noise from data
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Q4. Explain K-Fold Cross-Validation.

Answer:

K-Fold CV splits the dataset into **k equal parts (folds)**. The model is trained on k-1 folds and tested on the remaining fold. This repeats k times, each fold acting as the test set once.

Final performance = **average score across all k folds**.

Common choice: $k = 5$ or $k = 10$

Benefits:

- Every data point is used for both training and testing
- More reliable performance estimate than a single train/test split
- Reduces variance in evaluation

Stratified K-Fold is used for imbalanced classification – it preserves class distribution in each fold.

Q5. How do you handle imbalanced datasets?

Answer:

Imbalanced data = one class has far more samples than others (e.g., 95% negative, 5% positive in fraud detection).

Techniques:

Approach	Description
Oversampling	Duplicate minority class samples (SMOTE creates synthetic ones)
Undersampling	Remove majority class samples
Class weights	Assign higher penalty for misclassifying minority class (<code>class_weight='balanced'</code>)
Threshold tuning	Change decision threshold (default 0.5)
Use better metrics	Precision, Recall, F1, ROC-AUC instead of accuracy
Ensemble methods	BalancedBaggingClassifier, EasyEnsemble

Q6. Explain Precision, Recall, F1-Score, and ROC-AUC.

Answer:

Given a confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Metric	Formula	Meaning
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Overall correctness
Precision	$TP/(TP+FP)$	Of all predicted positives, how many are actually positive?
Recall (Sensitivity)	$TP/(TP+FN)$	Of all actual positives, how many did we catch?
F1-Score	$2 \cdot (P \cdot R) / (P + R)$	Harmonic mean of Precision and Recall
ROC-AUC	Area under ROC curve	Model's ability to distinguish between classes (1.0 = perfect)

When to use what:

- **Precision:** When FP is costly (spam detection – don't flag real mail)
- **Recall:** When FN is costly (cancer detection – don't miss a case)
- **F1:** When both FP and FN matter, and classes are imbalanced
- **ROC-AUC:** General ranking ability of classifier

Q7. What is regularization? Explain L1 vs L2.

Answer:

Regularization adds a penalty term to the loss function to prevent overfitting by discouraging large weights.

L1 Regularization (Lasso):

$$\text{Loss} = \text{MSE} + \lambda \cdot \sum |w_i|$$

- Produces **sparse models** (drives some weights exactly to 0)
- Acts as automatic **feature selection**
- Robust to outliers

L2 Regularization (Ridge):

$$\text{Loss} = \text{MSE} + \lambda \cdot \sum w_i^2$$

- Shrinks weights toward zero but **never exactly zero**

- All features are retained
- Works better when all features are relevant

Elastic Net = L1 + L2 combination

λ (lambda) is the regularization strength. Higher $\lambda \rightarrow$ more regularization $\rightarrow$ simpler model.

Q8. What is gradient descent? Explain its variants.

Answer:

Gradient descent is an optimization algorithm that minimizes the loss function by iteratively updating parameters in the direction of the negative gradient.

$$w = w - \alpha \cdot \partial L / \partial w$$

Where α = learning rate.

Variant	Update Frequency	Pros	Cons
Batch GD	After seeing ALL data	Stable convergence	Slow on large data
Stochastic GD (SGD)	After EACH sample	Fast, can escape local minima	Noisy updates
Mini-Batch GD	After a small batch (e.g., 32)	Balanced; most common in DL	Requires tuning batch size

Common optimizers (DL): Adam, RMSprop, AdaGrad, Momentum SGD

Q9. What is the difference between a generative and discriminative model?

Answer:

	Generative Model	Discriminative Model
What it learns	Joint probability $P(X, Y)$	Conditional probability $P(Y X)$
Can generate data?	Yes	No
Examples	Naive Bayes, GANs, HMMs, VAEs	Logistic Regression, SVM, Neural Nets
Training data needed	More	Less

Q10. Explain the difference between bagging and boosting.

Answer:

Both are **ensemble methods** that combine multiple weak learners.

Bagging (Bootstrap Aggregating):

- Trains models **in parallel** on random bootstrap samples
- Combines by **averaging** (regression) or **voting** (classification)
- Reduces **variance**
- Example: **Random Forest**

Boosting:

- Trains models **sequentially**, each correcting the previous model's errors
- Combines by **weighted sum**
- Reduces **bias**
- Examples: **XGBoost, LightGBM, AdaBoost, Gradient Boosting**

Q11. What is a confusion matrix? What can you derive from it?

Answer:

A confusion matrix is a table showing the performance of a classification model:

	Predicted: Positive	Predicted: Negative
Actual: Positive	TP (True Pos)	FN (False Neg)
Actual: Negative	FP (False Pos)	TN (True Neg)

From it you can derive: Accuracy, Precision, Recall, F1-Score, Specificity ($TN/(TN+FP)$), False Positive Rate.

Q12. What is the difference between parametric and non-parametric models?

Answer:

	Parametric	Non-Parametric
Assumes	Fixed functional form	No fixed form
Parameters	Fixed number	Grows with data

	Parametric	Non-Parametric
Examples	Linear Regression, Logistic Regression, Naive Bayes	KNN, Decision Trees, SVM with RBF kernel
Memory	Compact (just parameters)	Requires storing data
Risk	Underfitting if wrong assumptions	Overfitting if not tuned

Q13. Explain how Random Forest works.

Answer:

Random Forest builds multiple Decision Trees using:

1. **Bootstrap sampling** – each tree is trained on a random subset of training data (with replacement)
2. **Feature randomness** – at each split, only a random subset of features is considered

Final prediction:

- **Classification:** majority vote across all trees
- **Regression:** average of all tree predictions

Hyperparameters to tune:

- `n_estimators` – number of trees
- `max_depth` – maximum depth of each tree
- `max_features` – features considered per split
- `min_samples_split` – minimum samples to split a node

Advantages: Robust to outliers, handles missing values, provides feature importance.

Q14. Explain how XGBoost works.

Answer:

XGBoost (Extreme Gradient Boosting) builds trees **sequentially**. Each new tree corrects residual errors from previous trees.

Key formula: At step t , the new model is:

$$F_t(x) = F_{t-1}(x) + \eta \cdot h_t(x)$$

Where η = learning rate, h_t = new tree.

XGBoost improvements over standard GBM:

- Uses **second-order gradients** (Newton's method) for faster convergence
- **Regularization** built-in (L1, L2)
- **Column subsampling** (like Random Forest)
- **Sparsity-aware** split finding (handles missing values)
- **Parallel tree construction** within each level

Q15. What is PCA (Principal Component Analysis)?

Answer:

PCA is a **dimensionality reduction** technique that transforms features into a new set of orthogonal components (principal components) that capture maximum variance.

Steps:

1. Standardize the data (zero mean, unit variance)
2. Compute the **covariance matrix**
3. Compute **eigenvectors and eigenvalues**
4. Sort by eigenvalue (largest = most variance)
5. Project data onto top-k eigenvectors

When to use: Remove multicollinearity, visualize high-dimensional data, speed up training, reduce noise.

Limitation: Components are linear combinations — interpretability is lost.

Q16. What is the difference between classification and regression?

Answer:

	Classification	Regression
Output	Discrete class labels	Continuous numeric value
Examples	Spam/Not spam, Sentiment	House price, Temperature
Common algorithms	Logistic Regression, SVM, RF, XGBoost	Linear Regression, Ridge, SVR, RF
Common metrics	Accuracy, F1, AUC	MSE, RMSE, MAE, R ²

Q17. What are the assumptions of Linear Regression?

Answer:

1. **Linearity** – relationship between X and Y is linear
2. **Independence** – observations are independent
3. **Homoscedasticity** – constant variance of residuals
4. **Normality** – residuals are normally distributed
5. **No multicollinearity** – features are not highly correlated

Loss function (OLS):

$$\text{MSE} = (1/n) \cdot \sum (y_i - \hat{y}_i)^2$$

Q18. What is Logistic Regression and how does it work?

Answer:

Logistic Regression is a **classification** algorithm (despite the name). It models the probability of a class using the **sigmoid function**:

$$\sigma(z) = 1 / (1 + e^{(-z)})$$

$$\text{where } z = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

Output is a probability between 0 and 1. Decision boundary at 0.5 by default.

Loss function (Binary Cross-Entropy / Log Loss):

$$L = -(1/n) \cdot \sum [y_i \cdot \log(\hat{p}_i) + (1-y_i) \cdot \log(1-\hat{p}_i)]$$

Q19. What is SVM (Support Vector Machine)?

Answer:

SVM finds the **optimal hyperplane** that maximizes the margin between classes. The data points closest to the hyperplane are called **support vectors**.

Hard margin SVM (linearly separable):

$$\begin{aligned} &\text{maximize: } 2/||w|| \quad (\text{maximize margin}) \\ &\text{subject to: } y_i(w^T x_i + b) \geq 1 \end{aligned}$$

Soft margin SVM (with slack variables ξ):

$$\text{minimize: } (1/2) ||w||^2 + C \cdot \sum \xi_i$$

C = regularization parameter. High C → less margin, fits training data more tightly.

Kernel trick: Maps data to higher dimensions to find non-linear boundaries.

- Linear kernel
- RBF (Gaussian) kernel: $K(x, z) = \exp(-\gamma ||x-z||^2)$
- Polynomial kernel

Q20. What is feature selection vs feature extraction?

Answer:

	Feature Selection	Feature Extraction
What it does	Selects a subset of original features	Creates new features from originals
Output	Subset of original features	Transformed/new features
Methods	Chi-squared, ANOVA F, RFE, Lasso	PCA, LDA, Autoencoders, Word2Vec
Interpretability	High (original features kept)	Lower (new abstract features)

Your research connection: In your Bangla sentiment analysis project, you used **Chi-squared (χ^2)** and **ANOVA F** for feature selection – this is a great example to cite!

Q21. What are evaluation metrics for regression?

Answer:

Metric	Formula	Notes
MAE	$(1/n) \cdot \sum y_i - \hat{y}_i $	Robust to outliers
MSE	$(1/n) \cdot \sum (y_i - \hat{y}_i)^2$	Penalizes large errors more
RMSE	$\sqrt{\text{MSE}}$	Same unit as target variable
R ²	$1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$	Proportion of variance explained (1 = perfect)
MAPE	$(1/n) \cdot \sum (y_i - \hat{y}_i) / y_i \cdot 100$	Percentage error; interpretable

Q22. What is the difference between supervised, unsupervised, and semi-supervised learning?

Answer:

	Supervised	Unsupervised	Semi-Supervised
Labels	Yes (labeled data)	No labels	Small labeled + large unlabeled
Goal	Predict labels	Discover structure	Leverage unlabeled data
Examples	Classification, Regression	Clustering, Dimensionality Reduction	Self-training, GANs
Algorithms	RF, SVM, Neural Nets	K-Means, DBSCAN, PCA	Label Propagation

Q23. Explain K-Means Clustering.

Answer:

K-Means partitions data into k clusters by minimizing within-cluster variance.

Algorithm:

1. Initialize k cluster centroids randomly
2. Assign each point to the nearest centroid (Euclidean distance)
3. Recompute centroids as mean of assigned points
4. Repeat steps 2-3 until convergence

Objective:

minimize: $\sum_i \sum_{x \in C_i} ||x - \mu_i||^2$

Limitations:

- Must specify k in advance
- Sensitive to outliers and initialization
- Assumes spherical clusters
- Struggles with non-convex shapes

Elbow method – plot inertia vs k; choose the "elbow" point.

Q24. What is data leakage and how do you prevent it?

Answer:

Data leakage occurs when information from the test set (or future) "leaks" into the training process, causing unrealistically good performance.

Types:

- **Target leakage:** A feature is derived from the target variable (e.g., using post-event data)
- **Train-test contamination:** Preprocessing (e.g., scaling, imputation) done on full dataset before splitting

Prevention:

- Always split data **before** any preprocessing
 - Use **Pipeline** objects in sklearn to apply transformations only to training folds
 - Scrutinize features that have suspiciously high correlations with the target
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Q25. What is hyperparameter tuning? What methods are used?

Answer:

Hyperparameters are settings that control the learning process (not learned from data). Examples: learning rate, n_estimators, max_depth, C (SVM).

Tuning methods:

Method	Description	Pros/Cons
Grid Search	Try all combinations in a defined grid	Exhaustive but slow
Random Search	Randomly sample from hyperparameter space	Faster, often as good
Bayesian Optimization	Uses past results to guide next search	Efficient, fewer evaluations
Halving/Successive Halving	Progressively eliminate poor configurations	Fast

```
from sklearn.model_selection import GridSearchCV, RandomizedSearchCV
```
